

ALT YOLO: Altitude Aware YOLO Maritime UAV Vision for Search and Rescue

Lydia Abraham
COE 595
Department of ICS,
King Fahad University of Petroleum and Minerals
Dammam, Saudi Arabia

Abstract—Maritime search and rescue heavily relies on unmanned aerial vehicles to provide a wide-area visual coverage. Object detection in the field of maritime has additional challenges compared to areal object detection for numerous reasons. The object of interests are typically very small, while the sea-surface reflection, background clutter, altitude of UAV, camera angle presents strong variation. This paper uses SeaDronesSee Object Detection V2 as primary benchmark and proposes three contribution to address these challenges.

The first contribution evaluates the official YOLO26-P2 architecture, which adds a stride-4 detection head for tiny objects. Two training configurations are compared: M3a, trained with box-loss weight 7.5, and M3b, trained with an increased box-loss weight of 9.0 (cls=0.5 in both configurations). M3a achieves mAP@50 of 0.7653 and M3b achieves 0.7524, both below the YOLO26n baseline (0.8150). M3b improves recall from 0.7287 to 0.7410, but mAP@50-95 decreases from 0.4282 to 0.4256. The 50-epoch training budget may be one factor in P2 underperformance, but these results do not establish its cause.

The second contribution is altitude-conditioned SAHI inference, which adapts tile size and overlap to per-frame altitude metadata (256×256 at high altitude, 320×320 at medium, 480×480 at low). SAHI-only inference (Condition C) achieves mAP@50 of 0.7319 with recall of 0.8557 on the validation split, compared with 0.8150 and 0.7854 for the baseline, respectively, indicating a recall gain at the cost of lower mAP and substantially greater latency. The combined SAHI and glare preprocessing pipeline (Condition D) produces an average of 10.54 detections per image versus 6.60 for SAHI alone, a 59.7% increase in detection count, but mAP@50 decreases further to 0.6239. A larger detection count does not by itself demonstrate improved accuracy.

Finally, the third contribution is a glare-aware selective enhancement pipeline using HSV-based specular reflection detection and CLAHE contrast restoration. Applied at inference time, glare preprocessing reduces overall mAP@50 by 9.18 percentage points (0.8150 to 0.7232). Glare coverage averages 1.5% across the 1,125 unlabelled test images, with only 26 images exceeding 10% coverage; these observations suggest that uniform preprocessing may be unnecessary for many images, but do not establish the cause of the accuracy decrease.

Per-altitude evaluation reveals a 23.53 percentage-point mAP@50 gap for the baseline between medium (0.8483) and high (0.6130) altitude, identifying altitude as an important observed performance factor without establishing it as the sole cause of detection failures.

Index Terms—maritime search and rescue, UAV, SeaDronesSee, small object detection, object detection, aerial imagery

1 INTRODUCTION

Search-and-rescue operations are highly time-sensitive, and rapid localization of people in distress can significantly affect mission outcomes. In this context, UAVs offer an attractive platform because they can cover large maritime areas more flexibly and at lower cost than crewed aircraft. Their practical value, however, depends not only on flight capability but also on the reliability of onboard or near-real-time visual perception systems that can detect rescue targets under challenging conditions.

Open-water UAV imagery presents a particularly difficult detection problem. Human targets often occupy only a very small portion of the frame due to flight altitude and camera perspective, while the sea surface introduces clutter through waves, glare, reflections, foam, and rapidly changing color patterns. As a result, maritime SAR detection combines two hard problems: small-object recognition and severe background interference. These challenges make direct transfer from general aerial or terrestrial object detection insufficient for robust maritime deployment.

SeaDronesSee addresses an important gap in this area by providing a dedicated benchmark for maritime UAV perception [1]. Beyond large-scale annotation, the dataset includes per-frame metadata such as altitude, camera angle, speed, and time, which enables more detailed analysis of when and why models fail. This is especially important in maritime settings, where changes in viewing geometry and environmental conditions can strongly affect visibility and scale. As a result, SeaDronesSee serves not only as a benchmark for performance comparison but also as a foundation for condition-aware modeling and evaluation.

Motivated by this benchmark, a growing body of work has proposed improved detectors for maritime and UAV imagery. Some studies modify mainstream one-stage detectors to better preserve fine-grained features for tiny targets [2], [3]. Others focus on training strategies for limited supervision [4], or on lightweight and real-time designs intended for embedded deployment [5], [6]. In parallel, broader reviews have summarized the state of maritime UAV detection and small-object detection more generally [7], [8]. Although these works collectively show progress, their evaluation settings, datasets, and deployment assumptions vary substantially, making direct comparison difficult.

The contributions of this paper are as follows. (1) We establish

a per-altitude evaluation protocol using SeaDronesSee metadata and observe a 23.53 percentage-point mAP@50 gap between medium and high altitude for the baseline. (2) We evaluate the official YOLO26-P2 architecture on maritime data under a controlled training budget and compare default and increased box-loss weights, alongside bounding-box size analysis. (3) We implement altitude-conditioned SAHI inference that adapts slice parameters to per-frame altitude metadata and compare its detection counts with fixed slicing; the tested adaptive settings do not increase average detection counts in the reported altitude-bin samples. (4) We implement and evaluate an HSV-based glare-aware preprocessing pipeline and characterise its effect on validation mAP. All experiments use a fixed random subset of SeaDronesSee ODv2 with seed=42. Ground truth annotations for the official test split are not publicly available; AP and recall results are reported on the validation subset, while unlabelled test images are used for glare-coverage statistics.

2 RELATED WORKS

2.1 Benchmark and Dataset Foundations

SeaDronesSee is the central reference point for recent research on maritime UAV vision for SAR [1]. Its main contribution is the introduction of a dedicated open-water benchmark with annotated maritime objects related to rescue scenarios, together with three evaluation tasks: object detection, single-object tracking, and multi-object tracking. Unlike many general-purpose aerial datasets, SeaDronesSee also records environmental and flight metadata such as altitude, camera angle, speed, and time. This makes it particularly valuable for studying how detection performance changes under different capture conditions rather than treating all frames as statistically similar.

The benchmark is especially important because maritime UAV imagery differs substantially from conventional aerial detection datasets. While datasets such as UAVDT and VisDrone provide useful evidence on small-object detection, scale variation, and motion-related degradation, they are primarily focused on urban or traffic scenes rather than open-water environments [1], [8]. Consequently, SeaDronesSee helps establish maritime SAR perception as a distinct research setting with its own background statistics, target appearance, and operational constraints.

2.2 Maritime-Specific Detector Adaptations

Several studies build directly on SeaDronesSee or closely related maritime scenarios by adapting established object detectors to the open-water setting. YOLOv7-sea is a representative example that targets two core failure modes in SeaDronesSee: very small swimmers and heavy sea-surface interference [2]. The model extends YOLOv7 by adding an additional small-object prediction head and integrating SimAM, a parameter-free attention mechanism intended to emphasize informative regions. In addition to network changes, the method also relies on practical strategies such as image cropping, test-time augmentation, and prediction fusion to improve benchmark performance.

These maritime-oriented adaptations reflect a common design assumption in the literature: open-water rescue targets require stronger preservation of high-resolution detail and better suppression of clutter than standard aerial detectors typically provide. This leads many methods to focus on multi-scale feature enhancement, local attention, and improved inference pipelines. Although these

approaches often report higher accuracy, their deployment relevance depends on whether the gains remain valid under realistic onboard constraints, particularly when large inputs and expensive post-processing are removed.

2.3 Tiny-Object Detection in UAV Imagery

A second line of related work comes from the broader UAV tiny-object detection literature. Improved YOLOv8n for UAV imagery is an example of a lightweight detector modified to better handle small targets through an additional detection head, enhanced feature fusion, attention mechanisms, and improved localization loss [3]. Although this work is not limited to maritime scenes, it is relevant because it addresses the same core technical difficulty: preserving weak visual evidence for tiny objects under aggressive downsampling.

More generally, tiny-object detection research shows that performance often depends on how well a detector retains fine spatial detail while maintaining sufficient semantic information [7]. Common solutions include multi-scale fusion, improved label assignment, attention modules, and refined regression losses. These ideas are highly relevant to swimmer detection in SeaDronesSee because rescue targets frequently appear at the limit of visibility. However, performance gains obtained on generic aerial benchmarks do not automatically translate to open-water imagery, where clutter characteristics and object-background contrast differ significantly.

2.4 Low-Shot Learning and Efficient Detection

Another relevant direction addresses the scarcity of labeled maritime data. Low-shot open-water monitoring investigates how transfer learning, dynamic augmentation, and improved regression loss can improve performance when only limited annotated data are available [4]. This is important for SAR applications because collecting and labeling rescue imagery is expensive and operationally constrained. Rather than relying only on large-scale supervised training, low-shot approaches aim to make better use of limited data and to improve generalization to related benchmarks such as SeaDronesSee.

Efficiency-oriented detection is also closely related to practical SAR deployment. YOLO-OC emphasizes real-time small-object detection in ocean environments by combining architectural refinement with re-parameterization to improve inference speed while maintaining accuracy [5]. Similarly, Bio-YOLO targets lightweight aerial small-object detection through a bio-inspired architecture designed to improve object-background separation without incurring a large parameter budget [6]. These works are relevant because maritime UAV systems must operate under limited onboard computation, power, and latency budgets. Even when such methods are not evaluated directly on SeaDronesSee, they provide useful evidence that accuracy and efficiency must be considered jointly rather than as separate goals.

2.5 Survey Literature and Remaining Gaps

Recent survey papers provide additional context for positioning SeaDronesSee within the broader literature. A general survey on small-object detection organizes the field around themes such as multi-scale feature fusion, anchor and assignment strategies, contextual semantics, attention mechanisms, and optimization objectives [7]. This taxonomy is useful for understanding why small targets remain challenging and for interpreting the kinds

of modifications proposed in detector design. However, it is not specific to maritime imagery and therefore does not fully capture challenges such as glare, water reflection, or the value of flight metadata.

A more domain-focused review of maritime UAV detection argues that maritime scenes should be treated as a distinct research problem rather than as a simple extension of general aerial detection [8]. It groups methods into categories such as scale-aware detection, small-object detection, view-aware methods, rotated detection, and lightweight approaches, while also reviewing relevant maritime datasets. Together, these survey papers show that current research is moving toward more specialized modeling for maritime UAV imagery. At the same time, they also reveal several gaps that remain central to this paper: inconsistent evaluation protocols across datasets, limited condition-aware analysis, and insufficient attention to embedded real-time deployment.

3 DATASETS

3.1 SeaDronesSee OD V2

This paper uses the SeaDronesSee Object Detection V2 dataset [9]. There are 14,227 images with 8930 training sets, 1547 validation and 3750 testing set. There are five classes namely swimmer, boat, jetski, buoy, and life saving appliance. The image resolutions of these datasets are 3840x2160, 5456x3632, and 1229x934 for ODV2. The benchmark also provides a frame level metadata such as altitude, camera angle, speed, and time, and evaluates detection using AP50:0.05:95, AP50, AP75, AR1, and AR10. This dataset was chosen since for two main reasons. First, it was dataset used on the main reference of this project. Additionally the dataset included per image metadata such as altitude, and camera angle. Due to computational constraints, all experiments are conducted in a fixed random subset of 4,268 images drawn from the full dataset with a fixed random seed (seed=42). The subset was drawn from the full collection and split as follows:

- Training: 2679 images (62.8%)
- Validation: 464 images (10.9%)
- Test: 1125 images (26.4%)

The original COCO JSON annotations contains an additional class called ignored with (category id = 0) marking ambiguous or occluded regions. This annotation is excluded before building the YOLO class map and the selected images were kept in a subset and label was generated from the JSON.

3.2 Reference Aerial Benchmarks

3.2.1 UAVDT and VisDrone (aerial benchmarks) [1]

UAVDT and VisDrone are common benchmark providing annotated objects in land-based areas and UAVs. In this survey they are used to compare aerial versus maritime unique features. They cover scale variation and motion blur common to UAV capture, but they do not emphasize per-frame environmental metadata in the way SeaDronesSee does [1]. In a SeaDronesSee-centered narrative, these datasets help motivate why tiny-object handling and feature fusion are necessary, while also highlighting that maritime detection introduces qualitatively different background statistics.

3.2.2 AI-TOD and TinyPerson (aerial tiny-object datasets) [6]

AI-TOD and TinyPerson are aerial benchmarks that stress tiny-object detection, commonly used to test whether a model can retain small-scale detail under heavy downsampling and clutter [6]. Bio-YOLO uses these datasets to show that a lightweight architecture can remain competitive while emphasizing object-background separation and efficiency [6]. In this survey, these datasets provide evidence that parameter-efficient design can still address tiny-object regimes, which is directly relevant to onboard UAV constraints.

3.3 Maritime and Ocean-Scene Benchmarks

3.3.1 SeaDronesSee

SeaDronesSee is an open-water maritime UAV benchmark designed around SAR detection and tracking. It provides annotated maritime objects related to people in water and supports three tasks: object detection, single-object tracking, and multi-object tracking [1]. A defining feature is the availability of per-frame environmental metadata such as altitude and camera angle (and additional flight/capture parameters), allowing models or analyses to condition on capture context rather than assuming a single homogeneous distribution [1].

3.3.2 CURPC 2022 (ocean/underwater-style benchmark) [5]

CURPC 2022 is used for ocean visual interference object detection targeting small objects. The focus on small targets, strong clutter, and domain-specific noise that can overwhelm weak object cues makes it relevant [5]. Although the sensing geometry may differ from UAV open-water scenes, CURPC is valuable for studying techniques that explicitly trade accuracy for speed and for motivating real-time evaluation practices.

3.3.3 Open-water low-shot dataset (used by [4])

The open-water low-shot dataset is used for evaluation and generalization testing in the SeaDronesSee [4]. Since it reflects practical search and rescue constraints, such as labeled maritime data being scarce, and strong models must adapt under limited supervision rather than relying only on large-scale annotation.

3.4 Dataset Comparison

4 METHODOLOGY

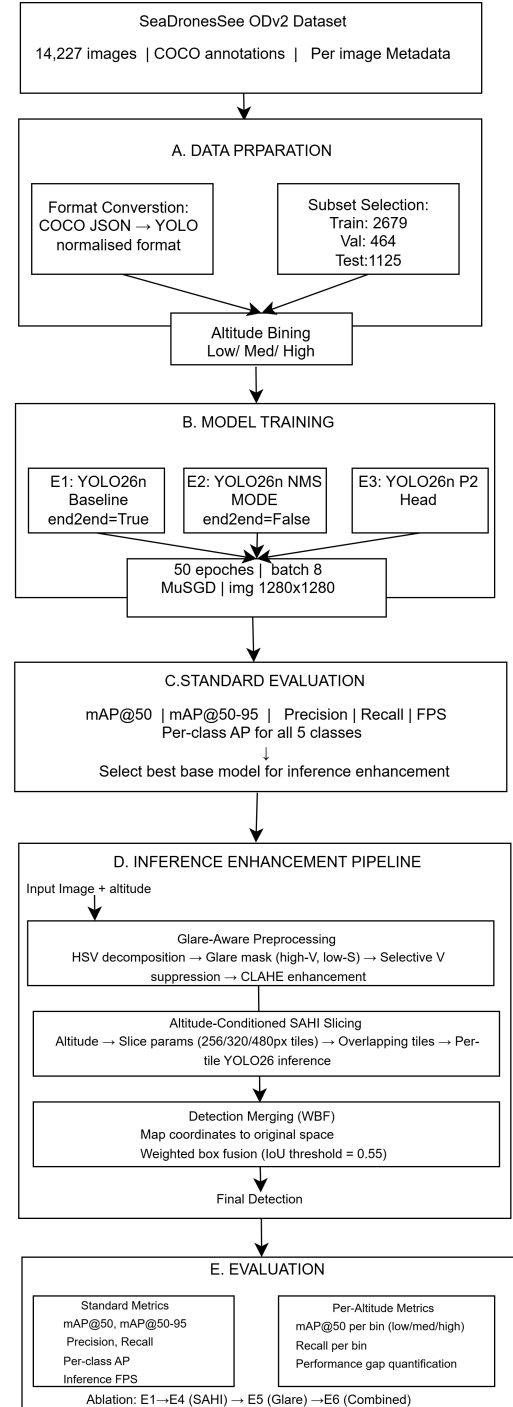
4.1 Overview

This work addresses three complementary challenges in open-water tiny-object detection: scale variation caused by UAV flight altitude, missed detection of sub-pixel-scale objects, and degraded feature quality in regions affected by sun glare. The methodology is organised around three contributions applied to YOLO26 as a base detector and evaluated on a fixed subset of the SeaDronesSee Object Detection V2 benchmark. Figure 1 illustrates the complete methodology framework. The pipeline consists of five stages: dataset preprocessing, P2 tiny head object detection implementation, design and evaluation of components independently using inference enhancement pipeline combining altitude conditioned SAHI and glare-aware preprocessing, and systematic per class and per altitude evaluation. Each contribution is first evaluated independently using its primary metric. Model architectures (M1,

TABLE 1: Dataset comparison across the reviewed works (reported statistics).

Dataset	Environment	Platform	Classes	Size	Resolution	Tasks
SeaDronesSee (original benchmark)	Open-water maritime	UAV	6 classes	5630 images; 230 videos; ~400K instances	3840×2160, 5456×3632	OD, SOT, MOT
UAVDT	Urban/traffic	UAV	Traffic objects	Not reported	Width 1024	OD/Tracking
VisDrone2022	Urban/traffic	UAV	10 classes	10K images; 400 videos; >2.6M instances	2000×1500; 3840×2160	OD
CURPC 2022	Underwater/ocean	Underwater dataset	4 classes	9000 training images	1920×1080	OD
AI-TOD	Aerial tiny objects	Aerial imagery	8 categories	28,036 images; 700,621 instances	Varied aerial images	OD
TinyPerson	Aerial coasts/beaches	Aerial imagery	2 classes	1610 images; 72,651 objects	497×700 to 4064×6354	OD
Open-water shot subset	low-maritime	UAV	3 classes	Subset of 2000 UAV images	Not reported	OD

M2, M3a, M3b) are compared using mAP@50 and mAP@50-95 on the validation split. SAHI inference is evaluated using both detection count analysis and a manual mAP computation against validation ground truth. Glare preprocessing is evaluated through a four-condition ablation (A: baseline; B: glare only; C: SAHI only; D: SAHI and glare combined) on the validation split. A combined pipeline evaluation reports the joint detection-count and mAP results when SAHI and glare preprocessing are applied together. All experiments use the same fixed data split (seed=42) and identical training configuration unless otherwise stated. Ground truth annotations for the test split are not publicly available; all quantitative results are therefore reported on the validation split.

**Fig. 1:** Methodology pipeline

4.2 Dataset Preparation

4.2.1 Source Dataset

The SeaDronesSee Object Detection V2 dataset has 14,227 UAV images of maritime open-water scenes, including resolutions of 3840 x 2160, 5456 x 3632, and 1229 x 934 pixels. The five scored detection classes are swimmer, boat, jet ski, life saving appliance, and buoy; the ignored category (id 0) is excluded from the training class map. Ground truth is provided in COCO JSON format with per-frame environmental metadata including UAV altitude, gimbal angle, and flight speed.

Due to constraints, a fixed random subset of approximately 30% of the dataset was used across experiments (seed=42). The selected subset uses the following split; preservation of the full dataset distribution has not been independently established:

- Training: 2,679 images (62.8 %)
- Validation: 464 images (10.9 %)
- Test: 1,125 images (26.4 %)
- Total: 4,268 images

4.2.2 Annotation Conversion

The original COCO JSON annotations are converted to YOLO normalised bounding box format. Each annotation is transformed as follows: a ground truth bounding box defined in COCO format as $[x_{\min}, y_{\min}, \text{width}_{\text{coco}}, \text{height}_{\text{coco}}]$ within an image of dimensions $W \times H$, the YOLO format label is computed as:

$$\begin{aligned} x_{\text{centre}} &= \frac{x_{\min} + \text{width}_{\text{coco}}/2}{W} \\ y_{\text{centre}} &= \frac{y_{\min} + \text{height}_{\text{coco}}/2}{H} \\ \text{width} &= \frac{\text{width}_{\text{coco}}}{W} \\ \text{height} &= \frac{\text{height}_{\text{coco}}}{H} \end{aligned} \quad (1)$$

All values are normalised to [0,1] to make the representation scale invariant. The conversion to YOLO is not only computationally fast, but also required by the Ultralytics training pipeline.

4.2.3 Class Distribution and Imbalance

The SeaDroneSee dataset has sever class imbalance. Swimmer instance dominate the training data, as main target of interest, while rare classes such as jet ski and buoy have fewer instances. The class loss is biased towards the majority class potentially causing the model to underweight rare classes. On the contrary the rare objects appears fewer times per epoch. Hence limiting generalisation. The YOLO26 baseline uses class-loss weight cls = 0.5, but the effect of this choice on class imbalance was not isolated.

4.2.4 Altitude Stratification for Evaluation

To analyze the relationship between performance and altitude, validation images are stratified into altitude bins using the metadata CSV:

- **Low altitude:** $h < 15$ metres (objects largest, detection easiest expected)
- **Medium altitude:** $15 \leq h < 40$ m (intermediate scale)
- **High altitude:** $h \geq 40$ metres (objects smallest, detection hardest expected)
- **Unknown:** no metadata available

The geometric relationship motivating these thresholds is:

$$\text{apparent_pixel_size} = \frac{\text{focal_length_px} \times \text{real_size_m}}{\text{altitude_m}} \quad (2)$$

For a swimmer of approximate dimensions 0.5m x 1.7m:

- **At altitude 10m:** apparent size $\approx 65 \times 90$ pixels (easily detectable)
- **At altitude 30m:** apparent size $\approx 22 \times 30$ pixels (challenging)
- **At altitude 80m:** apparent size $\approx 8 \times 11$ pixels (near detection limit)

4.3 Baseline Model Training

The baseline of this project is the latest YOLO series of real-time object detectors, YOLO26 (Ultralytics, 2025), which is optimized for edge and low-power devices [10].

4.3.1 YOLO26 Architecture

The architecture of YOLO26 has the following key features:

- **DFL Removal**
YOLO26 simplified inference by removing the Distribution Focal Loss (DFL) module.
- **End-to-End NMS-Free Inference**
YOLO26 is configured for end-to-end inference by default [10]. The exact matching implementation should be verified against the Ultralytics version used in these experiments; the following expression illustrates a generic classification-and-localisation matching cost, rather than a verified implementation detail:

$$C(i, j) = \lambda_{\text{cls}} \times L_{\text{cls}}(\hat{p}_i, y_j) + \lambda_{\text{box}} \times L_{\text{box}}(\hat{b}_i, b_j) \quad (3)$$

where:

- $\hat{p}_i$: predicted class probability vector for prediction i
- y_j : ground truth class one-hot vector for object j
- $\hat{b}_i$: predicted bounding box for prediction i
- b_j : ground truth bounding box for object j
- $\lambda_{\text{cls}}, \lambda_{\text{box}}$: weighting coefficients

- **ProgLoss + STAL**
ProgLoss and STAL are described as mechanisms intended to improve training for small objects [10]. Their individual effects and the exact assignment behaviour were not isolated or verified in the experiments reported here.
- **MuSGD Optimizer**
It has hybrid optimizer composing SDG and Muon [10].
- **Up to 43% Faster CPU Inference**
The cited model documentation reports faster CPU inference [10]; this work does not independently benchmark CPU or onboard UAV performance.

The diagram below shows the architecture of YOLO26 [11].

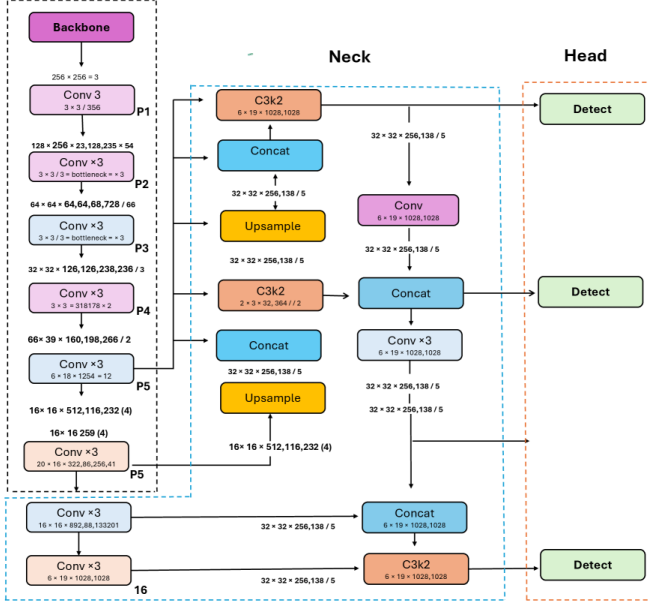


Fig. 2: YOLO 26 Architecture

4.3.2 Experiment 1 - Baseline YOLO26n (End to End)

```

Model:      yolo26n.pt (COCO pretrained,
               fine-tuned on subset)
Architecture: YOLO26n with 3 detection
               heads (P3, P4, P5)
               P3: stride 8 -> finer detail
                  / smaller objects
               P4: stride 16 -> intermediate
                  -scale objects
               P5: stride 32 -> coarser
                  features / larger objects
Input size: 1280 x 1280
Epochs:    50
Batch size: 8
Optimiser:  MuSGD (default YOLO26 config)
box loss:   7.5
cls loss:   0.5
end2end:    True (default NMS-free mode)

```

This baseline configuration represents the latest available detector applied to SeaDroneSee without architectural modification.

4.3.3 Experiment 2 - NMS YOLO26n

```

Model:      yolo26n.pt
Architecture: YOLO26n with 3 detection
               heads
               end2end: False (NMS re-
                  enabled)
box loss:    7.5
cls loss:    0.5 (same as baseline)
All other settings identical to Experiment 1

```

Disabling the end-to-end setting enables the NMS-based inference configuration used in Experiment 2. The class-loss weight is kept consistent for fair comparison.

Comparing these two configurations evaluates their observed detection performance; it does not, without implementation-level verification, isolate the effects of one-to-one versus many-to-one assignment.

4.4 Contribution 1: YOLO26n-P2 for Tiny Maritime Objects

4.4.1 Problem

The highest-resolution detection head in standard YOLO26n operates on P3 feature maps (80x80 for 640x640 input), corresponding to a stride of 8 pixels. Swimmers at high altitude can occupy only a few pixels. The following hypothetical reference-box comparison illustrates how IoU is sensitive to box size; it does not describe a verified YOLO26 anchor-assignment mechanism:

Example: ground truth box = 8 × 10 pixels

Illustrative reference box = 13 × 10 pixels

Intersection: 8 × 10 = 80 pixels²

Union: 13 × 10 + 8 × 10 − 80 = 130 + 80 − 80 = 130 pixels²

IoU = 80/130 = 0.615

In the illustrative reference-box comparison, IoU = 0.615. For a 5 × 6 pixel swimmer:

Intersection: 5 × 6 = 30 pixels²

Union: 130 + 30 − 30 = 130 pixels²

IoU = 30/130 = 0.231 (illustrative overlap)

A low IoU in this illustrative calculation does not by itself establish whether a prediction is assigned during YOLO26 training; the actual assignment rule must be checked in the implementation.

4.4.2 P2 Head Architecture

The P2 head adds a fourth detection branch operating on a 160×160 feature map (stride 4), obtained by upsampling the P3 features:

```

P3 features (160×160×256)
↓
Upsample 2x (320×320×256)
+
Backbone P2 (320×320×64) ← shallow backbone
features
↓
Concatenate (320×320×320)
↓
CBS block (320×320×128) [Conv + BatchNorm +
SiLU]
↓
Detection head output:
  stride-4 prediction features (exact tensor
  layout implementation-dependent)

```

This is intended to retain finer spatial detail than the P3 head; a four-pixel detection limit was not established by these experiments.

4.4.3 Training Configurations

Two training configurations are evaluated for the P2 architecture, corresponding to experiments M3a and M3b:

M3a — Default P2 (box=7.5):

```

Model:      official yolo26-p2.yaml
               initialised from yolo26n.pt
box loss:    7.5 # (M3a maritime-tuned P2)
cls loss:    0.5
imgsz: 1280
batch: 8
seed: 42
All other settings identical to baseline

```

M3b — Default P2 (box=9.0):

```
Model:      official yolo26-p2.yaml
            initialised from yolo26n.pt
box loss:   9 # (M3b maritime-tuned P2)
cls loss:   0.5
imgsz: 1280
batch: 8
seed: 42
All other settings identical to baseline
```

The P2 model is initialised from the COCO-pretrained YOLO26n weights. The new P2 head layers are randomly initialised and trained from scratch while the backbone and existing heads continue fine-tuning from pretrained weights.

4.4.4 Expected Vs Observed Outcome

Under the current training budget of 50 epochs on the subset, the P2 head underperforms the baseline. One possible explanation is that the newly initialised P2 layers require additional optimisation steps; another is that maritime object scales differ from the pretraining domain. Neither explanation is isolated by the current experiments, and the reported results do not establish an anchor-assignment mismatch in YOLO26. The experiment documents the observed effect of adding a tiny-object detection branch under the evaluated training constraints and motivates further controlled investigation.

4.5 Contribution 2: Altitude-Conditioned SAHI Inference

4.5.1 Problem

YOLO26 processes images at a fixed input resolution of 1280×1280 pixels. A 4K SeaDronesSee image (3840 × 2160) is downsampled to 1280×1280 during inference, compressing objects that were already tiny. A swimmer occupying 8 × 11 pixels in the original 4K image is compressed to approximately 2 × 3 pixels after this downscaling, placing it at or below the effective detection limit of the model.

SAHI (Slicing Aided Hyper Inference) [?] avoids this compression loss by processing overlapping image tiles at a resolution closer to the original, so that objects retain more pixels within each tile:

Standard inference:

```
3840x2160 image → downscale → 1280x1280 →
YOLO26 → detections
Swimmer: 8px → compressed to ~3px → near
detection limit
```

SAHI inference:

```
3840x2160 image → slice into overlapping
tiles →
  each tile passed to YOLO26 independently
→
  merge detections back to original
coordinates → candidate detections
```

Scope of evaluation. SAHI is evaluated through detection counts and inference time, as well as a manual mAP calculation against the 464-image validation subset with available annotations. Official hidden-test benchmark evaluation was not performed. The manual validation estimates should not be assumed identical to official server scores without protocol verification.

4.5.2 Mathematical Formulation of Slicing

For an image of size $H \times W$ with slice dimensions $s_h \times s_w$ and overlap ratios o_h and o_w , the step sizes are:

$$\begin{aligned}\Delta h &= s_h \times (1 - o_h) \\ \Delta w &= s_w \times (1 - o_w)\end{aligned}\tag{4}$$

The number of slices in each dimension is:

$$\begin{aligned}n_h &= \left\lceil \frac{H - s_h}{\Delta h} \right\rceil + 1 \\ n_w &= \left\lceil \frac{W - s_w}{\Delta w} \right\rceil + 1\end{aligned}\tag{5}$$

The total number of inference passes is:

$$\text{Total slices} = n_h \times n_w\tag{6}$$

Example for a 640 × 640 image with $s = 320$, $o = 0.2$:

$$\begin{aligned}\Delta &= 320 \times (1 - 0.2) = 256 \\ n &= \lceil (640 - 320)/256 \rceil + 1 = 2 + 1 = 3 \\ \text{Total: } &3 \times 3 = 9 \text{ slices}\end{aligned}$$

The overlap can reduce the likelihood that objects near tile boundaries are truncated in every tile, but it does not guarantee complete coverage or correct localisation.

4.5.3 Detection Merging

After running inference on all tiles, detected boxes are mapped back to the original image coordinate space. In this work, SAHI is configured to use **Greedy NMM** (Non-Maximum Merging) as the postprocessing strategy, with an IoU match threshold of 0.5. This approach suppresses duplicate detections produced by the same object appearing in multiple overlapping tiles while retaining distinct detections.

4.5.4 Altitude-Conditioned Slicing Strategy

Standard SAHI applies fixed slice parameters to all images regardless of capture conditions. The proposed framework adapts slice size and overlap to UAV flight altitude using the per-frame metadata provided by SeaDronesSee. The geometric rationale is: object apparent size scales inversely with altitude, therefore finer slicing is needed at higher altitudes to ensure tiny objects are adequately represented within each tile.

There are three altitude bins and their corresponding slice parameters used in the experiment as summarized in Table 2.

TABLE 2: Altitude-conditioned SAHI slice parameters.

Altitude Bin	Threshold	Slice Size	Overlap	Rationale
high	≥ 40 m	256 × 256	0.30	Objects <10 px; maximise tile coverage
medium	15–40 m	320 × 320	0.20	Moderate objects; standard slicing
low	< 15 m	480 × 480	0.10	Large objects; avoid unnecessary splits
unknown	—	320 × 320	0.20	Conservative default

The altitude bin for each image is derived from the `height_above_takeoff(meter)` field in the SeaDronesSee COCO JSON metadata. Images with missing or null altitude values are assigned to the unknown bin and receive the default parameters. **Inference cost trade-off.** Each slice requires one full YOLO26 forward pass. Smaller slices produce more tiles and therefore higher total inference latency. At high altitude the smaller slices are intended to improve tiny-object coverage, but the measured

inference latency limits their suitability for real-time deployment. At low altitude, objects are sufficiently large that standard full-image inference is adequate, and larger tiles minimise redundant processing.

4.5.5 Scope of Evaluation

SAHI inference is evaluated at two levels. First, a detection-count comparison (fixed 320×320 slicing vs. altitude-conditioned slicing) is reported for the M1 baseline on a 20-image sample per altitude bin drawn from the validation split. Second, a full mAP evaluation is conducted on the complete validation set (464 images) using a manual precision-recall computation: SAHI predictions are saved in YOLO label format and matched against ground truth annotations using IoU thresholding at 0.50:0.05:0.95. This produces manual validation-set AP estimates; equivalence to the official COCO evaluation implementation has not been established.

4.6 Contribution 3: Glare Aware Selective Enhancement

4.6.1 Problem

Reflection from the sun on water bodies create bright regions in UAV images leading to:

- **Overwhelm local contrast** in their vicinity, obscuring nearby objects
- **Confuse the detector feature extraction** such bright reflection is perceived as edge during feature extraction it activate edge and blob detectors similarly to real objects
- **Create false positive detections**, classes as such buoy and life-saving appliances which are intrinsically bright such reflection increases the false positive rate.

Standard image normalisation does not address this because glare is spatially non-uniform it depends on sun angle, wave state, and UAV viewpoint.

4.6.2 Glare Detection Algorithm

According to specular reflection in physics, glare is image region that has simultaneously high-brightness and low saturation. Water reflects sunlight as white or near white light with no chromatic information rather than high luminance.

In HSV colour space:

- **H (hue):** colour angle 0°–360°
- **S (saturation):** colour purity 0–255 (0 = grey/white, 255 = fully saturated)
- **V (value):** brightness 0–255

The glare mask $G(x, y)$ is computed as:

$$G(x, y) = \begin{cases} 1 & \text{if } V(x, y) > \tau_V \text{ and } S(x, y) < \tau_S \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

Where:

- τ_V : brightness threshold (typically 210–230)
- τ_S : saturation threshold (typically 30–50)

A life jacket (bright orange) has: $V \approx 240$ (bright), $S \approx 200$ (highly saturated) $\rightarrow G = 0$, not flagged as glare. A glare patch: $V \approx 250$ (very bright), $S \approx 15$ (near-white) $\rightarrow G = 1$, correctly flagged.

4.6.3 Glare Suppression and Local Enhancement

The processing pipeline applies two operations in sequence:

Operation 1 — Selective intensity suppression:

The output intensity is computed as:

$$V_{\text{out}}(x, y) = V_{\text{in}}(x, y) \times (1 - G(x, y) \times (1 - \beta)) \quad (8)$$

Where $\beta \in [0, 1]$ is the suppression factor ($\beta = 0.6$ means 40% reduction).

At non-glare pixels ($G = 0$): $V_{\text{out}} = V_{\text{in}} \times (1 - 0) = V_{\text{in}}$ (unchanged)

At glare pixels ($G = 1$): $V_{\text{out}} = V_{\text{in}} \times \beta = V_{\text{in}} \times 0.6$ (40% reduction)

Operation 2 CLAHE local contrast enhancement:

CLAHE (Contrast Limited Adaptive Histogram Equalisation) use histogram to map accurately how light is distributed in the image time mathematically. The algorithm breaks down image in small local grids and equalize them dependently without affecting the rest of the frame. It enhances regions both low and high contrast.

CLAHE algorithm:

- 1) Divide image into tiles of size $T \times T$ (typically 8×8).
- 2) For each tile:
 - a) Compute histogram $h(k)$ of intensity values.
 - b) Clip histogram at limit L to prevent over-amplification:
 $h_{\text{clip}}(k) = \min(h(k), L)$
 Redistribute excess count uniformly across all bins.
 - c) Compute cumulative distribution function (CDF).
 - d) Apply CDF as intensity mapping to tile pixels.
- 3) Apply bilinear interpolation at tile boundaries to prevent blocking artefacts.

The clip limit L controls the trade off between enhancement and noise amplification. While too high over amplifies a noise in homogeneous regions such as open water, too low provides minimal enhancement. $L = 3.0$ is used as determined by visual inspection on a validation sample.

Mathematical formulation of CLAHE mapping:

For a tile with clipped histogram h_{clip} , the intensity mapping function $T(v)$ is:

$$\text{CDF}(k) = \frac{\sum_{j=0}^k h_{\text{clip}}(j)}{N_{\text{tile}}} \quad (9)$$

$$T(v) = (N_{\text{intensity}} - 1) \times \text{CDF}(v) \quad (10)$$

Where:

- N_{tile} : total pixels in the tile
- $N_{\text{intensity}}$: number of intensity levels (256 for 8-bit)
- v : input intensity value
- $T(v)$: output intensity value

4.6.4 Inference-Time Application

Glare preprocessing is applied at inference time for two major reasons. First, applying at training time not only would require preprocessing the whole dataset but also compromises important training variations. Second this application allows fair comparison to fairly compare model with and without preprocessing step.

4.7 Evaluation Protocol

4.7.1 Standard Metrics

All models are evaluated on the same validation split using the following performance metrics:

mAP@50: Mean average precision at IoU threshold 0.50 computed as the mean of per-class AP:

$$AP_c = \int_0^1 P_c(R) dR \approx \text{area under precision-recall curve for class } c \quad (11)$$

$$mAP@50 = \frac{1}{C} \times \sum_c AP_c \quad (12)$$

Where C = number of classes = 5.

mAP@50:95: COCO-style mAP averaged over IoU thresholds 0.50, 0.55, 0.60, ..., 0.95:

$$mAP@50:95 = \frac{1}{10} \times \sum_t mAP@t \quad (13)$$

$$t \in \{0.50, 0.55, \dots, 0.95\}$$

This metric rewards precise localisation more strongly than mAP@50 and is particularly revealing for tiny object detection where even small coordinate errors significantly reduce IoU.

Precision and Recall:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (14)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (15)$$

Where:

- **TP:** detected object with IoU ≥ 0.50 with a ground truth box
- **FP:** detected object with IoU < 0.50 or duplicate detection
- **FN:** ground truth object with no detection having IoU ≥ 0.50

4.7.2 Per-Altitude Evaluation

Each model is evaluated with four altitude bins namely high, medium, low and unknown. Per-altitude mAP@50 and Recall are key metrics used to evaluate whether improvements are generalized across capture condition or concentrated in specific altitude region.

4.7.3 Per-Class Analysis

Swimmers and buoy AP are separately reported since both are affected by scale variation (for swimmers) and glare (for buoy).

4.7.4 Inference Speed

For all models inference speed is measure by milliseconds per image. SAHI configurations report per image time including slicing overhead to enable realistic deployment.

4.8 Combined Pipeline: SAHI and Glare Preprocessing

A combined inference pipeline applies altitude-conditioned SAHI and glare-aware preprocessing jointly. Glare preprocessing is applied to each image before SAHI slicing: the HSV glare mask is computed, glare regions are suppressed, CLAHE contrast enhancement is applied, and the resulting image is passed to SAHI for tiled inference. This pipeline corresponds to Condition D in the ablation study. The combined pipeline is evaluated on the full validation set (464 images) using both average detection count per image and the manual mAP computation described in Section 4.5.5. The combined mAP evaluation is not equivalent to a full end-to-end validation against a held-out labelled test set; it is an approximation using the validation split and a manual matching procedure. All claims about combined pipeline performance should be interpreted accordingly.

4.9 Experimental Comparison Table

TABLE 3: Experimental setup for YOLO26n models and components.

Exp	Model	E2E	SAHI	Glare	Primary Metric	Purpose
E1	M1 – YOLO26n baseline	✓	✗	✗	mAP@50 (val)	Primary baseline
E2	M2 – YOLO26n + NMS	✗	✗	✗	mAP@50 (val)	Strategy comparison
E3	M3a – YOLO26-P2 def.	✓	✗	✗	mAP@50 (val)	Ablation: P2 head
E4	M3b – YOLO26-P2 tuned	✓	✗	✗	mAP@50 (val)	Anchor/Loss tuning
E5	M1 + fixed SAHI	—	✓	✗	Det count	Fixed-slice baseline
E6	M1 + alt-conditioned	—	✓	✗	Det count, mAP	Metadata-aware inf.
E7	M1 + glare prep	—	✗	✓	mAP@50 (val)	Glare effect study
E8	M1 + SAHI + glare	—	✓	✓	Det count, mAP	Pipeline evaluation

5 EXPERIMENTAL SETUP REVIEW

5.1 Input Resolution and Preprocessing

The images used for all the experiments has **1280 X 1280 pixels** to preserve fine spatial details critical for detecting small maritime objects. The preprocessing steps include 4 stages:

- Conversion of COCO to YOLO format
- Removal of the ignored category
- Verification of Label Integrity
- Dataset consistency checks

Inaddition, glare aware analysis was conducted using HSV based characteristics to identify specular refelection regions. This preprocessing was not intagrated in to model however utilized during inference.

5.2 Training Configuration

To ensure fair comparison, all models were trained under a controlled and consistent configuration.

- Model family: YOLO26
- Epochs: 50
- Batch size: 8
- Input size: 1280
- Optimizer settings: default Ultralytics configuration
- Data split: fixed (seed = 42)
- Augmentation: mosaic enabled, mixup disabled

Three model variants were evalutaed under this experiment:

- M1 (Baseline): Standard YOLO26n with default detection head
- M2 (NMS variant): YOLO26n with non-end-to-end detection (end2end=False)

- M3 (P2 variant): YOLO26 with an additional stride-4 detection head for small objects

For Model 3: Official P2 YOLO26n model specific maritime tuning was applied to improve performance. Anchor box computation was based on K-means clustering on dataset bounding boxes. To increase localization loss weight was given as box = 9.0 .

5.3 Hardware and Inference Protocols

All experiments were executed on a GPU enabled environment across multiple Paperspace Gradient notebook sessions. Training experiments (M1, M2, M3) were executed on an NVIDIA Quadro P6000 GPU (24 GB VRAM). Inference and ablation evaluations were executed on an NVIDIA Quadro RTX 4000 GPU (8 GB VRAM). All environments used Python 3.11.7, PyTorch 2.1.1–2.6.0, and Ultralytics 8.4.41.(NVIDIA RTX 4000).

Two inference protocols were utilized:

- 1) Standard Inference: full image detection using trained YOLO models
- 2) SAHI based inference: slicing images into overlapping tiles and merging predictions. Under SAHI configuration there were two variants namely:
 - Fixed slicing: used default 320X320 tiles with contrast overlap
 - Altitude conditioned slicing: tiles size adapt based on altitude metadata

6 RESULTS REVIEW

6.1 Main Model Comparison

The quantitative result for the three model configurations evaluated on the validation split of SeaDronesSee subset is presented as follows. Table 4 presents quantitative results for all four model configurations evaluated on the validation split. M1 and M2 converge to identical performance with the best checkpoint reached at epoch 35 for both. The identical metrics are confirmed by distinct MD5 hashes for the two model weight files and different args.yaml configurations (end2end: null vs end2end: false), ruling out any duplication error. This establishes identical reported aggregate metrics for M1 and M2 under the evaluated setup, but does not isolate the assignment mechanism or establish model capacity and training-data quantity as the dominant causes of error.

TABLE 4: Model Performance Comparison (Validation Split)

Model	mAP@50	mAP@50-95	Precision	Recall
M1 – YOLO26n Baseline	0.8150	0.4758	0.8610	0.7854
M2 – YOLO26n + NMS	0.8150	0.4758	0.8610	0.7854
M3a – YOLO26-P2 default (box=7.5)	0.7653	0.4282	0.7863	0.7287
M3b – YOLO26-P2 maritime-tuned (box=9.0)	0.7524	0.4256	0.7609	0.7410

6.2 Per-Class Performance

Life saving appliances class is consistently hardest to detect across all models. This could be due to high class imbalance. On the contrary, boat class achieves the highest AP across all models.

TABLE 5: Per-Class AP@50 on Validation Split

Model	Swimmer	Boat	Jetski	Life Saving Appliance	Buoy
M1 Baseline	0.8158	0.9583	0.8555	0.5497	0.8955
M2 NMS	0.8158	0.9583	0.8555	0.5497	0.8955
M3 P2 (box=7.5)	0.7825	0.9179	0.8119	0.4268	0.8874

6.3 Inference Speed

All three reported standard-inference model configurations run below 30 ms per image on the evaluation GPU, corresponding to more than 65 images per second under the measured conditions. These GPU timings do not establish real-time onboard UAV performance, which also depends on hardware, power, and pipeline overhead. The P2 model incurs modest inference overhead over the baseline.

TABLE 6: Inference Speed (ms per image, validation split)

Model	Prep.	Infer.	Post.	Total
M1	4.6	7.8	0.3	12.7
M2	5.9	8.3	0.2	14.4
M3a (P2)	4.5	10.5	0.2	15.2

6.4 SAHI Detection Count Analysis

SAHI was evaluated on the full validation set (464 images) using altitude-conditioned slice parameters and the M1 baseline model. Two evaluation modes were used: a detection-count comparison between fixed (320×320) and altitude-conditioned slicing across a 20-image sample per altitude bin, and a full manual mAP evaluation over all 464 validation images.

TABLE 7: Detection Count Comparison (20 images per altitude bin, M1)

Altitude Bin	Slice (fixed)	Avg Dets (fixed)	Slice (conditioned)	Avg Dets (conditioned)	Delta
high	320×320	8.30	256×256	8.20	-0.10
medium	320×320	6.65	320×320	6.65	0.00
low	320×320	2.05	480×480	1.95	-0.10
unknown	320×320	6.90	320×320	6.90	0.00

The altitude-conditioned approach does not increase average detection counts relative to fixed-parameter SAHI in this sample. At high altitude it produces 8.20 detections per image versus 8.30 with fixed slicing; at low altitude, 1.95 versus 2.05. The medium and unknown bins are unchanged. Detection counts alone cannot establish recall or localisation improvements, and the latency trade-off requires separate measurement.

TABLE 8: Full mAP Evaluation (464 validation images, M1, altitude-conditioned SAHI)

Condition	mAP@50	mAP@50-95	Precision	Recall	Avg Dets/Image
A: Baseline (no preprocessing)	0.8150	0.4758	0.8610	0.7854	—
C: SAHI only	0.7319	—	—	0.8557	6.60
B: Glare preprocessing only	0.7232	0.4053	0.7771	0.6890	—
D: SAHI + glare	0.6239	—	—	0.7862	10.54

Condition C (SAHI without glare) achieves recall of 0.8557, higher than the baseline (0.7854), while mAP@50 decreases from 0.8150 to 0.7319. This indicates a recall–mAP trade-off under the reported manual validation evaluation. The available results do not isolate whether the mAP decrease arises from false positives, duplicate merging, localisation errors, or other factors. The combined SAHI and glare preprocessing (Condition D) produces an average of 10.54 detections per image versus 6.60 for SAHI alone, a 59.7% increase in detection count. However, mAP@50 decreases further to 0.6239, and recall drops to 0.7862. The increase in detection count should not be interpreted as a true-positive gain; the combined pipeline does not improve overall mAP and does not preserve the SAHI-only recall gain. Inference latency for SAHI is substantially higher than standard inference. Processing 464 validation images

under Condition C required approximately 2,447 seconds (roughly 5.3 seconds per image), compared to approximately 12.7 ms per image for the baseline. This trade-off is acceptable for post-flight analysis but prohibitive for real-time onboard deployment.

6.5 Glare Analysis

Glare preprocessing (Condition B) reduces mAP@50 by 9.2 percentage points relative to unprocessed baseline. Glare statistics computed on the test images show a mean glare fraction of only 1.5% of pixels per image, with only 26 out of 1,125 images exceeding 10% glare coverage.

In a separate detection-count analysis, altitude-conditioned SAHI alone produces an average of 6.32 detections per image. Combined with glare preprocessing, this increases to 9.97 detections per image, a 57.8% increase. This is a comparison of SAHI with and without glare preprocessing, not a comparison of altitude-conditioned versus fixed slicing. The increased count cannot be interpreted as improved recall or recovery of additional true objects without matching predictions to ground truth.

TABLE 9: Glare preprocessing ablation (val split, M1 baseline)

Condition	mAP@50	mAP@50-95	Precision	Recall	AP50 swimmer	AP50 buoy
A: Baseline (no preprocessing)	0.8150	0.4758	0.8610	0.7854	0.8158	0.8955
B: + Glare preprocessing	0.7232	0.4053	0.7771	0.6890	0.6843	0.8286

6.6 Per-Altitude Evaluation

TABLE 10: Per-altitude mAP@50 on validation split

Model	High	Low	Medium	Unknown
M1-Baseline	0.6130	0.6673	0.8483	0.7032
M2-NMS	0.6074	0.6583	0.8593	0.7296
M3-P2	0.6124	0.6158	0.7879	0.6795

TABLE 11: Per-altitude Recall on validation split

Model	High	Low	Medium	Unknown
M1-Baseline	0.598	0.622	0.826	0.677
M2-NMS	0.591	0.653	0.862	0.751
M3-P2	0.558	0.623	0.756	0.647

Three takeaways from this experiment is note worthy. First, medium altitude consistently achieves the highest mAP@50 across all models. On the contrary, low altitude was expected to produce such results. This can be due to either high class enumeration or low altitude images contains cluttered foreground with partial occlusions leading to high false positive. Second, high-altitude performance is weakest across the reported models. This indicates an altitude-associated performance gap in the evaluated subset, but not a strictly monotonic relationship or a causal effect of altitude alone. Under the illustrative imaging assumptions used in Section 4.2.4, a swimmer at high altitude may occupy approximately 8×11 pixels in a 4K frame, which can become only a few pixels after resizing for inference. This illustrates a plausible scale challenge rather than a measured detection limit. Finally, NMS model (M2) shows improved recall over baseline at medium and unknown altitude (+3.6 and +7.4 percentage points, respectively). These bin-specific recall differences do not by themselves establish which assignment mechanism is responsible.

6.7 Discussion

The experimental results reveal three key findings. First, M1 and M2 achieve identical aggregate validation metrics under the evaluated configurations; this does not establish the assignment mechanism or the dominant source of errors. Second, the P2 detection head underperforms the baseline under the current training budget; additional training and model-specific scale analysis are possible directions for further testing, not demonstrated explanations. Third, life_saving_appliances has the lowest reported detection AP among the models in Table 5; class imbalance and visual similarity are possible contributors but were not isolated experimentally. The SAHI and glare pipelines were implemented and evaluated on the validation subset, including manual AP estimates. Neither improves overall mAP@50 relative to the baseline in the reported ablations. Official hidden-test evaluation remains future work.

6.8 SAHI Detection Count Analysis

7 SUGGESTED ENHANCEMENTS

7.1 Condition-Aware Evaluation as Standard Practice

SeaDronesSee provides per-frame altitude and angle metadata for every image. The per-altitude evaluation protocol introduced in this work should be adopted as a standard reporting requirement for future benchmark submissions. Reporting only aggregate mAP conceals systematic failure modes caused by altitude variation. Future work should extend this evaluation to include camera angle bins and temporal conditions such as time of day.

7.2 Maritime-Specific Anchor Calibration for P2 Head

The maritime object size distribution may differ from the pretraining domain, but the current experiments do not establish that the YOLO26-P2 implementation uses COCO-derived anchor boxes or that anchor mismatch explains its performance. Future work should use the available k-means bounding-box analysis to investigate whether model-specific scale handling or P2 configuration changes improve tiny-object detection. Any improvement in assignment efficiency or mAP must be demonstrated experimentally.

7.3 Extended Training and Data Augmentation for P2

The 50-epoch training budget on 2,679 images may be insufficient for the newly initialised P2 layers to reach their best performance, but convergence was not established by the reported comparison. Future experiments should use the full SeaDronesSee training set (8,930 images) with extended training to 100 or more epochs, combined with copy-paste augmentation that specifically synthesises additional small object instances.

7.4 Standardized SAHI Evaluation

SAHI has already been evaluated with a manual AP calculation on the 464-image labelled validation subset, in addition to detection counts and timing. Future work should verify the manual AP computation against a standard evaluator, compare fixed and altitude-conditioned SAHI using the same AP protocol, and submit predictions to the official hidden-test evaluation server. Increased detection counts alone do not establish true-positive recovery.

7.5 Glare-Stratified Evaluation

The glare pipeline has already been compared with the baseline using mAP on the labelled validation subset. Future work should stratify this comparison by measured glare coverage and examine per-class false positives to determine whether selective preprocessing helps specific conditions or categories. Official hidden-test evaluation remains outstanding.

7.6 Multispectral Fusion

SeaDronesSee includes multispectral footage. Fusing RGB with near-infrared or red-edge bands could improve human-water separability and reduce false positives caused by glare. This is a natural extension of the glare suppression contribution since NIR channels are less susceptible to specular reflection.

8 CONCLUSION AND FUTURE WORK

This work evaluated YOLO26-based detection on a maritime UAV search-and-rescue benchmark under the reported model configurations and two inference-time enhancements. The baseline YOLO26n achieves mAP@50 of 0.8150 on the SeaDronesSee validation subset, with medium-altitude performance higher (0.8483) than high-altitude performance (0.6130), a 23.53 percentage-point gap. This highlights altitude as an important observed factor, although the experiments do not establish it as the sole cause of detection failures. The NMS configuration produces identical aggregate validation metrics to the baseline, while the P2 configurations underperform under the evaluated 50-epoch budget. The available results do not establish why the P2 models underperform. Altitude-conditioned SAHI achieves higher recall than standard full-image inference (0.8557 versus 0.7854), but lower mAP@50 (0.7319 versus 0.8150) and substantially higher latency. The comparison with fixed slicing does not show increased average detection counts from altitude conditioning. Combining SAHI with glare preprocessing increases detection counts but reduces mAP@50 to 0.6239; the increased count is not evidence of more true-positive detections. Glare preprocessing alone also reduces mAP@50, from 0.8150 to 0.7232. These findings illustrate that additional architectural and inference complexity does not necessarily improve maritime detection accuracy. Future work should investigate longer P2 training and model-specific scale handling, validate the manual SAHI AP computation against a standard evaluator, compare fixed and adaptive SAHI using the same AP protocol, test more selective glare enhancement, and evaluate the final inference pipeline through the official SeaDronesSee hidden-test server.

REFERENCES

- [1] L. A. Varga, B. Kiefer, M. Messmer, and A. Zell, "Seadronessee: A maritime benchmark for detecting humans in open water," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022.
- [2] H. Zhao, H. Zhang, and Y. Zhao, "Yolov7-sea: Object detection of maritime uav images based on improved yolov7," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, 2023.
- [3] C. Sun, Y. Hu, and X. Jin, "Tiny object detection from uav viewpoint based on improved yolov8n," in *Proceedings of the 9th International Conference on Image, Vision and Computing (ICIVC)*, 2024.
- [4] W. Liao, Q. Ma, H. Wang, and H. Li, "Improving small object detection in open water maritime monitoring under low-shot learning," *Ocean Engineering*, 2026.
- [5] Y. Zhang, T. Qin, and Y. Zhou, "A yolo-oc real-time small object detection in ocean scenes," *Computer Vision and Image Understanding*, vol. 264, p. 104625, 2026.
- [6] J. Peng, S. Huang, C. Lin, and X. Pang, "Bio-yolo: A bio-inspired lightweight small object detection network for aerial images," *Applied Soft Computing*, vol. 190, p. 114575, 2026.
- [7] H. Wang and P. Gao, "Survey of small object detection methods based on deep learning," in *2024 9th International Conference on Intelligent Informatics and Biomedical Sciences (ICIIBMS)*, 2024.
- [8] C. Zhao *et al.*, "Deep learning-based object detection in maritime unmanned aerial vehicle imagery: Review and experimental comparisons," *Engineering Applications of Artificial Intelligence*, vol. 128, p. 107513, 2024.
- [9] University of Tübingen, "Seadronessee object detection v2 dataset," <https://cloud.cs.uni-tuebingen.de/index.php/s/ZZxX65FGnQ8zjBP>, 2024, accessed: 2024-05-20.
- [10] G. Jocher and J. Qiu, "Ultralytics yolo26," 2026. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [11] R. Sapkota *et al.*, "Yolo26: Key architectural enhancements and performance benchmarking for real-time object detection," 2025. [Online]. Available: <https://arxiv.org/abs/2509.25164>